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B.E. - (Computer) (Sem-VII) (Revised Course 2019-2020)
EXAMINATION DECEMBER 2023
Data Analytics

[Time: 3:00 Hours]**[Max. Marks: 100]****Instructions:**

- 1) Assume suitable data if necessary.
- 2) Answer any two questions from Part A and Part B. Also attempt one question from Part C.
- 3) Draw neat diagrams if required.
- 4) Write question numbers legibly while answering.
- 5) Write description for the questions based on the marks allotted.
- 6) Use of statistical tables permitted if required.

Part A

- Q1 A) What is the difference between Sampling and Re-sampling Methods. Discuss the features of Resampling methods. (05)
- B) Distinguish between supervised learning and unsupervised learning methods in data analytics. (05)
- C) Highlight the important characteristics of Big Data used in Analytics. (05)
- D) Explain the necessity of combining several algorithms for accomplishing a particular task. (05)
- Q2 A) Explain briefly ANOVA Test. (06)
- B) Calculate the Chi-square value for the following data of incidences of water-borne diseases in three tropical regions. (08)

	India	Equador	South America	Total
Typhoid	31	14	45	90
Cholera	2	5	53	60
Diarrhoea	53	45	2	100
	86	64	100	250

- C) On an examination given to the students at a large no of different schools, the (06)
mean grade was 74.5 and standard deviation 8.0. At one particular school, where
200 students took examination, the mean grade was 75.9. Discuss the significance
of the result at the 0.05 level from the view point of

- (i) One tailed test
(ii) Two tailed test.

- Q3 A) Given the dataset {a, b, c, d, e} and the following distance matrix given table 1, (10)
construct a dendrogram by complete linkage hierarchical clustering using the
agglomerative method.

	a	b	c	d	e
A	0	9	3	6	11
B	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

Table 1. Distance Matrix

- B) Determine the regression equation by finding the regression slope coefficient (06)
and intercept value using the following data.

X	55	60	65	70	75
Y	52	54	56	58	62

- C) Explain the following.

- i) Boosting
ii) Bagging

(04)

Part B

- Q4 A) What are Factors in R- Language? How are they created? How are they (06)
modified? Explain with R language code.
- B) Explain briefly different types database languages and their significance in (06)
DBMS.

- C) Give the comparison of different types of NOSQL databases with reference to their performances.

(08)

- Q5 A) Describe most commonly used data plots for statistical analysis in R Language.

(06)

- B) Explain Different phases of data analytic life cycle.

(06)

- C) Tabulate the features of SQL and NOSQL databases.

(08)

- Q6 A) With reference to data frames in R language.

(06)

- i) Create a simple data frame from 3 vectors. Order the entire data frame by the first column.

- ii) Create a data frame from a matrix of your choice, change the row names so every row says id_i (where i is the row number) and change the column names to variable_i (where i is the column number). i.e., for column 1 it will say variable_1, and for row 2 will say id_2 and so on.

- B) Point out the functions which are used for the conversion of covariance to correlation in R. Explain the function with syntax and sample data.

(06)

- C) Summarize the operations that can be performed on a data frame in R language.

(08)

PART C

- Q7 A) Explain with Step-by-Step procedure for the evaluation of machine language models in R language.

(10)

- B) State Bayes Theorem and illustrate with an example. How do we use numeric features in naïve Bayes algorithm?

(10)

- Q8 A) What do you, mean by information gain and entropy? How is it used to build the decision trees? Illustrate with an example.

(10)

- B) Explain in detail about Aprori Algorithm.

(10)